Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/22/2005
PI Name:	Shapiro, Irwin Ph.D.		
Project Title:	Solar-system tests of gravitational theories		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ishapiro@cfa.harvard.edu	**Fax:**	FY 617-495-7093
PI Organization Type:	UNIVERSITY	**Phone:**	617-495-7300
Organization Name:	Harvard University		
PI Address 1:	Smithsonian Astrophysical Observatory		
PI Address 2:	60 Garden Street, MS51		
PI Web Page:	http://cfa-www.harvard.edu/		
City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	01/01/2002	**End Date:**	12/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chandler, John (Smithsonian Astrophysical Observatory) Ratner, Michael (Smithsonian Astrophysical Observatory) Reasenberg, Robert (Smithsonian Astrophysical Observatory)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We are engaged in testing gravitational theory, mainly using observations of objects in the solar system and mainly on the interplanetary scale. For this project, we are analyzing a combination of observational data with our model of the solar system, including primarily planetary radar ranging, lunar laser ranging, and spacecraft tracking, but also including both pulsar timing and pulsar VLBI measurements. Our goal is either to detect departures from the standard model (general relativity) -- if any exist within the level of sensitivity of our data -- or to place tighter bounds on such departures. Einstein's general relativity is the accepted standard theory of gravitation. It serves as the basis of our understanding of the large-scale behavior of the universe, from the motions of planets in our own solar system to the time-evolution of the universal expansion, to the existence and properties of black holes. It is important, therefore, to ascertain the limits of validity for the theory, and this can be achieved only through observation of the effects of gravitation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We are engaged in testing graviational theory, mainly using observations of objects in the solar system and mainly on the interplanetary scale. Our goal is either to detect departures from the standard model (general relativity) - if any exist withing the level of sensitivity of our data - or to support this model by placing tighter bounds on any departure from it. For this project, 2we have analyzed a combination of observational data with our model of the solar system, including planetary radar ranging, lunar laser ranging, and spacecraft tracking, as well as pulsar timing and pulsar VLBI measurements. In the past year, we have added to our data, primarlily lunar laser ranging measuremtnyts, but also supplementary data concerning the physical properties of solar-system objects, such as the solar quadrapole moment, planetary masses, and asteroid radii. Because the solar quadrupole moment contributes to the classical precession of planetary perihelia, but with a dependence on distance from teh Sun that differs from that of the relativistic precession, it is possible to estimate effects simultaneously. However, our interest is maily in the relativistic effect, and we find that imposing a constraint on the quadrupole moment from helioseismology studies, gives us a dramatic (about ten-fold) decrease in the standard error of our estimate of the relativistic component of the perihelion advance. In addtion, by considering the contribution of the metric parameters beta and gamma to the signature of an equivalence principle violation, we have obtained another dramatic (nearly two orders of magnitude) reduction in the standard error for beta. This is the final year under grant NAG5-11455, but we have submitted a proposal to continue this research. In the coming year, we plan to continue adding data to our set, as available. We expect to use these data and improve models to obtain estimates of the graviational-theory parameters and to publish these results.
Bibliography Type:	Description: (Last Updated: 07/09/2007)